MURANG'A UNIVERSITY OF TECHNOLOGY

DEPARTMENT OF INFORMATION TECHNOLOGY

Unit code: SIT 305. **Unit Title:** HUMAN COMPUTER INTERACTION.

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NAME: Maina Mwangi.

TAKE AWAY CAT1

1. **What problems do users with cognitive impairments and learning difficulties face and what could be taken into consideration to ensure that your design supports users with those impairments.**

Problems

- **Information Overload:** Difficulty processing large amounts of information or complex layouts. This can lead to confusion and frustration.
- **Navigation Difficulties:** Struggling to understand navigational structures and finding relevant information. This can make it hard to complete tasks or achieve goals.
- **Memory Challenges:** Difficulty remembering instructions, steps, or information presented on the interface. This can hinder task completion and lead to errors.
- **Language and Comprehension:** Difficulty understanding complex language, jargon, or abstract concepts. This can limit comprehension and create barriers to access.

Design Considerations:

- **Clear and Concise Content:** Use simple language, avoid jargon, and break down information into manageable chunks.
- **Consistent Navigation:** Use clear and consistent navigation menus, breadcrumbs, and labels. Provide multiple ways to find information (e.g., search bar, sitemap).

- **Personalization and Control:** Allow users to customize the interface (e.g., font size, color contrast) to meet their individual needs.
- **Provide Clear Feedback:** Give users clear and immediate feedback on their actions to help them understand the system's response.
- **Accessibility Testing:** Conduct usability testing with users with cognitive impairments to identify and address any potential barriers.

2. **James has put forward the idea that thinking is divided into skill, rule and knowledge based thought. Explain these terms outlining what type of errors they can give rise to in a computer based system. (3 Marks)**

1. Skill-Based Errors:

- **Explanation:** These occur when highly practiced actions are performed unconsciously, like typing or driving a familiar route. It's about automatic, routine tasks.
- **Errors in Computer Systems:**
    - **Slips:** Unintended actions, like clicking the wrong button or mistyping a command.
    - **Lapses:** Memory failures, like forgetting to save a file or skipping a step in a process.

2. Rule-Based Errors:

- **Explanation:** When we encounter a familiar situation, we apply learned rules or procedures. This is more conscious than skill-based, but still relies on pre-existing knowledge.
- **Errors in Computer Systems:**
    - **Misapplication of good rules:** Applying a rule that's usually correct in the wrong context. For example, using a keyboard shortcut meant for one program in another.
    - **Application of bad rules:** Following incorrect or outdated procedures. For example, using an old password that's no longer valid.

3. Knowledge-Based Errors:

- **Explanation:** This is when we encounter a new or unfamiliar situation and have to problem solve using our knowledge and reasoning. It requires the most conscious effort.

- **Errors in Computer Systems:**
    - **Incorrect mental model:** Misunderstanding how the system works. For example, assuming a file is saved automatically when it's not.
    - **Biased reasoning:** Making decisions based on incomplete or inaccurate information. For example, trusting a phishing email because it looks official.

3. **Shneiderman (1998) devised Eight Golden Rules for interface design. List and explain 6 of these rules, give an explanation of what the rule is. (3 Marks)**

- **Strive for Consistency:** Use the same design patterns, visual language, and terminology throughout the interface. This makes it easier for users to learn and predict how the system will behave.
- **Enable Frequent Users to Use Shortcuts:** Provide shortcuts, hotkeys, or advanced features for experienced users to increase their efficiency.
- **Offer Informative Feedback:** Provide clear and immediate feedback to users about their actions and the system's response.
- **Design Dialog to Yield Closure:** Organize interactions into clear beginnings, middles, and ends. Provide users with a sense of accomplishment and clear next steps.
- **Offer Simple Error Handling:** Design the system to prevent errors whenever possible. When errors occur, provide clear and helpful error messages that guide users towards a solution.
- **Permit Easy Reversal of Actions:** Allow users to easily undo actions or revert to previous states. This reduces anxiety and encourages exploration.